Safety Enhanced UAV path planning using Sperical vector-based Particle Swarm Optimisation

Group name: s2_18_wc

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Project Overview and Objective

In this project, we are addressing the problem of planning a safe and most feasible path for an Unmanned Aerial Vehicle (UAV) which operates in a three-dimensional (3-D) environment which contains obstacles and non-uniform ground topology. Limitations in environmental information, sensing accuracy, and system representation create uncertainty that plays a critical role in determining safe and optimal UAV trajectories.

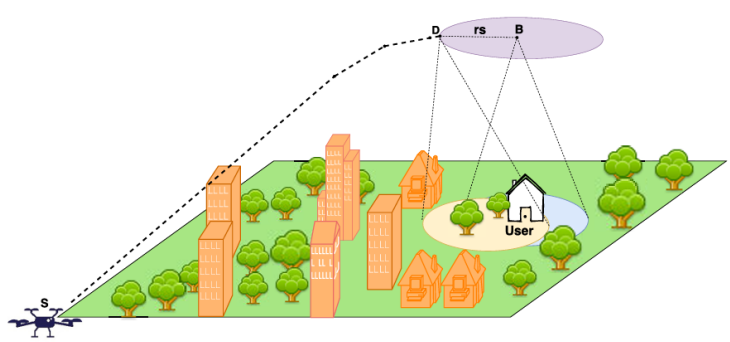

Figure 1: llustration of UAV navigating through obstacles and threat regions during path planning.

Our project aims at calculating an optimal UAV flight path that minimizes a cost function which is a combination of multiple criteria and meets safety and feasibility requirements. These limitations are obstacle evasion, altitude, motion smoothness, turning and climbing angle. The optimization is carried out in the form of a particle swarm optimization (PSO)-based algorithm,

that is, the Spherical Vector-based PSO (SPSO). This helps us to enable an efficient exploration of the configuration of path space [1].

The main sources of uncertainty include the terrains that are imperfect and the information of the obstacles in the path, error in measurements in UAV position and altitude, and simplifying the modeling in UAV dynamics. these are the uncertainties that motivate us to use the population based and randomized optimization techniques to obtain effective solutions.

Key Random Variables and Uncertainty Modeling

Various important random variables define the UAV path planning system, as a combination of these variables encloses the uncertainty due to the environmental factors, sensor constraints, and randomness inherent in the optimization. They are also very important variables in the modeling of the behavior of the system and determining the safety and efficiency of the generated flight paths.

- X_i: This represents the random variable which represents the i-th candidate encoded in UAV path as a waypoints sequence in 3D space.
- d_k: The distance between the path in which the UAV is travelling and the obstacle k, and this captures the uncertainty in the regions that influence the obstacle.
- θ and ϕ: Random variables denoting the turning and climbing/dive angles of the UAV, subject to feasibility constraints.
- $F(X_i)$: A random variable representing the evaluated cost of a candidate path, which is inclusive of path length, safety, and smoothness.

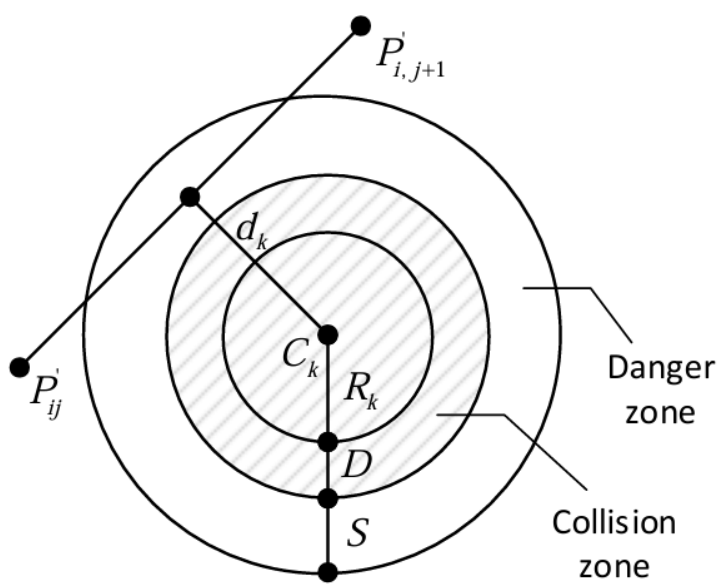

Figure 2: Cylindrical threat model used to represent obstacle danger zones in UAV path planning.

Uncertainty is implicitly represented with the stochastic sampling on the candidate paths in the PSO population, randomly generated velocity and position updates. Even though the swarm-based sampling procedure does not introduce explicit probability-density or mass functionalities,

the sampling procedure empirically induces a distribution on candidate paths. Stationary obstacles, bounded certainty that are absorbed by safety margins, and simplified UAV dynamics are included in modeling assumptions that we are using currently.

Probabilistic Reasoning and Dependencies

The result of this system in probabilistic reasoning on interaction of the candidate UAV path, restrictions by enviornment, and the evaluation mechanism of cost. The cost $F(X_i)$ is a function of cost on several uncertain conditions, such as ranges to obstacles, changes in the elevation of the terrain, and limitations due to motion capabilities of UAVs, including turn limits and climb limits. As a result, the cost of a path is evaluated to indicate effectiveness of a path as well as its vulnerability to possible danger. [2]

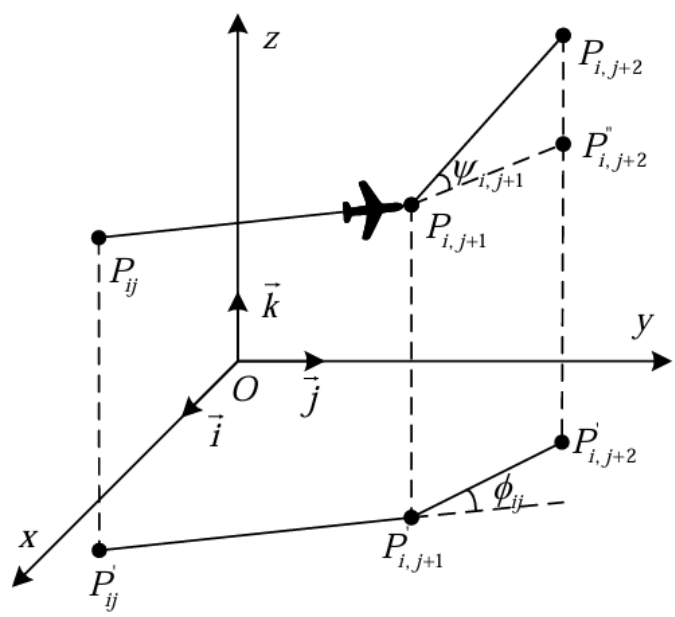

Figure 3: Turning and climbing angle constraints influencing UAV trajectory smoothness and feasibility.

In the Particle Swarm Optimization (PSO) algorithm, candidate path particles are assumed to be independent during the first stage of the algorithm which is known as the initialization stage. Dependencies arise naturally as the optimization goes on due to the social and self-learning aspects of PSO wherein the particles revise their path to rely on personal experience combined with information disseminated by the swarm. These interactions bring in conditional dependencies which lead the search to safer and more viable parts of the solution space.

The conditional reasoning is also used in the case of evaluating the feasibility of paths in that the cost attributed to a candidate path because it is determined by the satisfaction or failure of safety, obstacle avoidance, and motion constraints. The algorithm is able to obtain approximate predictions of the performance of the various solutions in case of uncertainty through repeated sampling, evaluation, and refinement of candidate paths through multiple iterations. The final decision making is that which minimizes the average cost, hence supporting sound and dependable

decision-making in uncertain operating conditions.

Model–Implementation Alignment

The probabilistic model is quite appropriate in the implementation of the project using the Spherical Vector-based Particle Swarm Optimization (SPSO) algorithm. During the implementation, every particle within the swarm is one of the possible UAV flight routes consisting of waypoints. Randomness is one of the major elements of the model and present in implementation through initialization and stochastic update rules.

Trajectories are randomly planted and particles move around in the 3 dimensional space by stochastic update equations because random numbers are used to get and explore the solutions. This shows the probabilistic fact that good directions cannot be exactly sampled. Simultaneously, the system is deterministic in its rules to provide safety and ability. As an example, constant safety margin, boundary on obstacles and UAV restrictions are enforced during the calculations of every path. These assumptions are made to make the environment simple with UAV being safe to operate. The cost functional determines the worth of every sampled path with respect to path length, collision removal, smoothness and energy viability, which is the same aspect as the safety and objective metrics displayed in the concept map.

The algorithm is conducted in various simulations (Monte Carlo simulations) to enhance reliability and represent the uncertainty. In a nutshell, probabilistic model determines the method used to explore the paths whereas implementation follows up the practical assumptions to assess and choose the optimal safe UAV path.

Cross-Milestone Consistency and Change

The probabilistic model at the present stage of the project is applied as randomized optimization involving the use of the Spherical Vector-based Particle Swarm Optimization (SPSO) algorithm. The model aims at searching a huge number of potential routes taken by the UAV and choosing the most optimal one depending on safety and cost. Some of the fundamentals of the model are already good and stable with previous milestones. A 3D terrain map that contains known obstacle and threat areas is used to model the environment of UAVs.

All candidate paths are characterized by a sequence of waypoints and path analysis is done in terms of a fixed cost function comprising of path length, collision avoidance, smoothness, and energy feasibility. Safety limits, e.g., constant safety margins on obstacles are also well defined and enforced. These factors have not changed since it is their basis of safe UAV navigation. Certain features of the probabilistic model will change as new milestones are made in the future. Presently, the uncertainties that affect environmental and measurement (i.e. gps error or terrestrial approximation) are treated in a simplistic manner with conservative assumptions. [3]

These uncertainties can be represented in later stages in a more explicit way with the help of probability distributions. Also, the surrounding is largely comfortable at this level, but the further activity will cover dynamic disturbances, including wind and moving barriers. More sophisticated statistical tests such as repetitive Monte Carlo simulation, and comparative tests are also going to be used to more rigorously test performance. These modifications are scheduled to be made since the

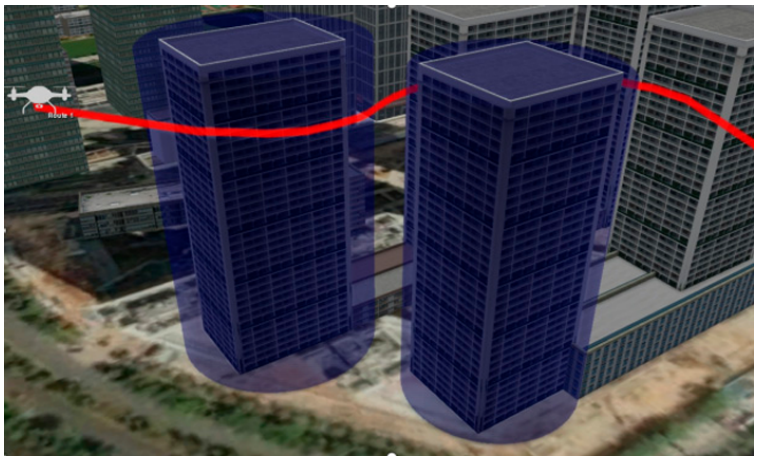

Figure 4: UAV flying between the obstacles, here, two building, after detection

existing assumptions are retained to provide the model with a computational simplicity and early development aptness. Subsequently, with the maturity of the project, a more realistic uncertainty and dynamics added will enhance the precision and consistency of the UAV path planning system.

Open Issues and Responsibility Attribution

The probabilistic problems remain unaddressed and at the present project phase, some of them are still unsolved. There s an important problem which is unsolved that is the need to model the uncertainties in the enviornment which includes the distribution of the obstacles which are unknownand error in elevation in terrains appropriately. These uncertainties are currently managed through constant safety margins which might be too safe. Premature convergence in the SPSO algorithm is also another open issue. Due to the randomized exploration implemented by the swarm, there are occasions when the swarm may end up reaching a local optimum solution instead of the global optimum.

This implies that there is a need of better exploration strategies. It is also unclear on the way the dynamic disturbances, including the wind or changing environment, are to be taken into consideration in the probabilistic model. These have not been explicitly modeled and is still subject to research and implementation as an open question. The theory and modeling group will have the role of enhancing the representation of uncertainty in the project. This involves improved management of factors such as unpredictable interferences, lack of GPS positioning, and the estimation of the terrain. The team will do the work of enhancing the SPSO algorithm and conducting additional simulations to test its functionality.

Another role that we will undertake is ensuring that the algorithm does not find itself entangled in bad solutions, which is achieved through the addition of additional exploration techniques. The following milestone will involve both teams collaborating and adding dynamic effects, i.e. the wind, or the environment, and testing the new model.

References

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